

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js automation engine** for MCA filings, reducing manual filing time by **85%** through reliable browser execution, retries, and API-backed state checks.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker**, deployed to **Google Cloud Run**, and automated releases through **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions** with PyTorch/TinyGrad model support and integration into an existing open-source application.
- Reduced runtime by **87%** using asynchronous and parallel processing, turning slow model workflows into a more usable developer/user experience.
- Improved 3D surface visualization around model outputs, making generated artifacts easier to inspect and validate.

Open Source

[Contributions](#) 

- Contributed to Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, adapting patches to established code review, issue context, and maintainer expectations.

Projects

Seaweed: Contest Platform | *Go, React, PostgreSQL, Judge0, Redis, AWS, Docker, Kubernetes*

[GitHub](#) 

- Architected a full-stack contest platform supporting **500+ concurrent users**, automated judging, live rankings, and feedback loops for programming contest workflows.
- Designed a sandboxed Judge0 pipeline with async Go dispatch, Docker/Kubernetes deployment on AWS, **sub-5s P95 verdict latency**, per-test telemetry, and PostgreSQL-backed result analysis.

Penny Lane Capital (Tradeage) | *Python, Agno, Pydantic, SQLite, yfinance, JSONL*

- Built an **Agno Workflow** multi-agent research system with analyst roles, bull/bear debate, trader, risk review, and portfolio approval using Pydantic state, checkpoints, memory, and JSONL traces.
- Evaluated agent decisions with historical-market backtests, reward loops, validation harnesses, and **5 baselines**, producing return, drawdown, and win-rate scorecards across market windows.

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Built an **AI gateway** for LLM, HTTP tool, and MCP traffic with route profiles, runtime policy checks, request scoring, and replayable decisions before downstream execution.
- Instrumented provider/tool flows with **OpenTelemetry** traces, Prometheus metrics, audit ledgers, and attack-lab validation so latency, failures, and unsafe actions could be investigated per request.

Technical Skills

Languages: Go, Python, TypeScript

Frameworks: FastAPI, Echo, Node.js, React

Tools: Judge0, PostgreSQL, Redis, Docker, GitHub Actions, OpenTelemetry, Prometheus